Laplacian Operator and Smoothing

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Smoothing as (mean curvature) Flow

- Model smoothing as a diffusion process

$$\boxed{\frac{\partial \mathbf{p}}{\partial t} = \lambda \Delta \mathbf{p}} = -2\lambda H \mathbf{n}$$

- Backward Euler for unconditional stability

$$\frac{\mathbf{p}^{(n+1)} - \mathbf{p}^{(n)}}{dt} = \lambda L \mathbf{p}^{\boxed{(n+1)}}$$

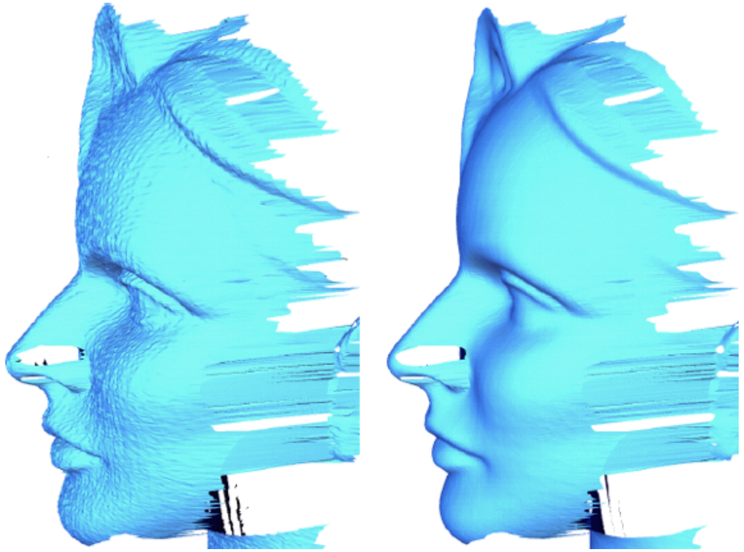
$$\mathbf{p}^{(n+1)} - \mathbf{p}^{(n)} = dt\, \lambda L \mathbf{p}^{(n+1)}$$

$$\boxed{(I - dt\, \lambda L)\mathbf{p}^{(n+1)} = \mathbf{p}^{(n)}}$$

Implicit Fairing: Implicit Euler Steps

- In each iteration, solve for the smoothed $\tilde{\mathbf{p}}$:

$$(I - \lambda L)\tilde{\mathbf{p}} = \mathbf{p}$$



Implicit fairing of irregular meshes using diffusion and curvature flow
M. Desbrun, M. Meyer, P. Schroeder, A. Barr, **ACM SIGGRAPH 99**

On meshes: smoothing as mean curvature flow

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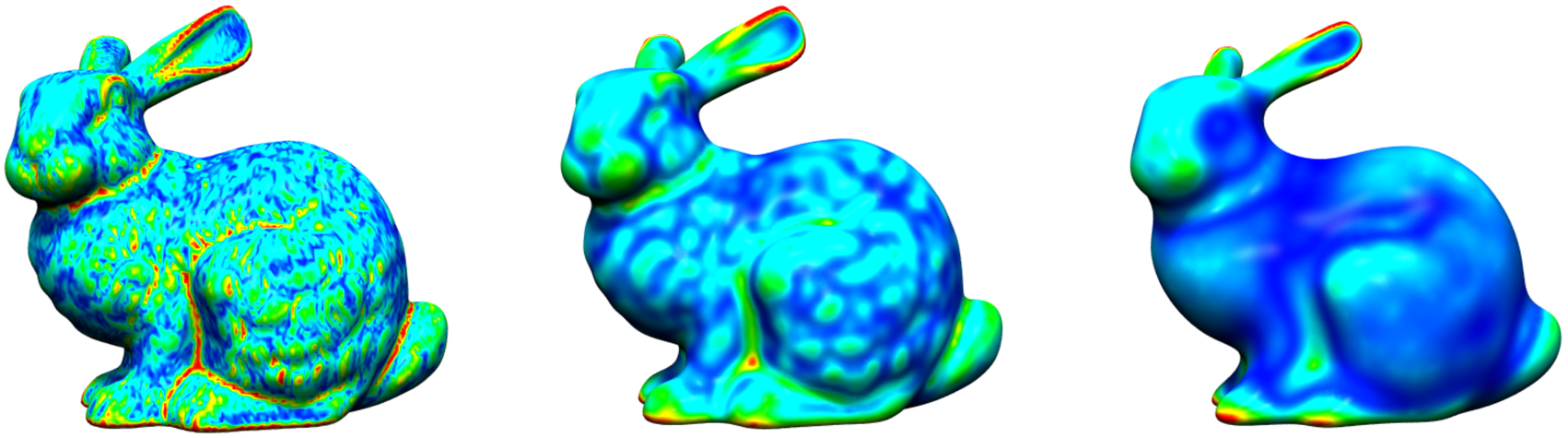
$$\boxed{\mathbf{p}^{(n+1)} = (I + dt\, \lambda L)\mathbf{p}^{(n)}}$$

Explicit integration! Unstable unless time step dt is small

Example



0 iterations

10 iterations

100 iterations

Smoothing as optimization

Minimizing a smoothness energy

- Let's go for $H = 0$

$$\Delta_{\mathcal{M}} \mathbf{p} = -2H\mathbf{n}$$

goal: $H = 0$ or $H = \text{const}$

- only trivial solution, no connection to initial surface p

$$\Delta_{\mathcal{M}} \tilde{\mathbf{p}} = 0$$

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- Let's regularize!

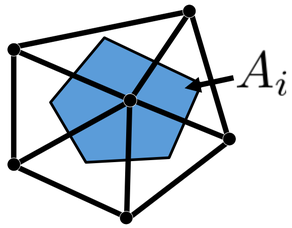
$$\Delta_{\mathcal{M}} \tilde{\mathbf{p}} = 0$$

weighting factor

$$\min_{\tilde{\mathbf{p}}} \int_{\mathcal{M}} \underbrace{\|\Delta_{\mathcal{M}} \tilde{\mathbf{p}}\|^2}_{\text{small } H} + w \underbrace{\|\tilde{\mathbf{p}} - \mathbf{p}\|^2}_{\text{stay close to original surface}}$$

Minimizing a smoothness energy

- Discretize: $$\min_{\tilde{\mathbf{p}}} \int_{\mathcal{M}} \|\Delta_{\mathcal{M}} \tilde{\mathbf{p}}\|^2 + w\|\tilde{\mathbf{p}} - \mathbf{p}\|^2$$



$$\min_{\tilde{\mathbf{p}}} \sum_{i=1}^{n} A_i \left(\|L\tilde{\mathbf{p}}_i\|^2 + w\|\tilde{\mathbf{p}}_i - \mathbf{p}_i\|^2 \right)$$

- Minimize!

$$\frac{\partial}{\partial \tilde{\mathbf{p}}} = 0$$

remember: $$\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = (A + A^T)\mathbf{x}$$

tip: google “The Matrix Cookbook”

Minimizing a smoothness energy

$$\min_{\tilde{\mathbf{p}}} \sum_{i=1}^{n} A_i \left(\|L\tilde{\mathbf{p}}_i\|^2 + w\|\tilde{\mathbf{p}}_i - \mathbf{p}_i\|^2 \right)$$

$$\tilde{\mathbf{p}} = [\tilde{\mathbf{x}}\ \tilde{\mathbf{y}}\ \tilde{\mathbf{z}}] = \begin{pmatrix} \tilde{p}_{1x} & \tilde{p}_{1y} & \tilde{p}_{1z} \\ \tilde{p}_{2x} & \tilde{p}_{2y} & \tilde{p}_{2z} \\ \vdots & \vdots & \vdots \\ \tilde{p}_{nx} & \tilde{p}_{ny} & \tilde{p}_{nz} \end{pmatrix} \in \mathbb{R}^{n\times 3}$$

$$E(\tilde{\mathbf{p}}) = (L\tilde{\mathbf{p}})^T M (L\tilde{\mathbf{p}}) + w(\tilde{\mathbf{p}} - \mathbf{p})^T M (\tilde{\mathbf{p}} - \mathbf{p})$$

$$\frac{\partial E}{\partial \tilde{\mathbf{p}}} = 2L^T M L\tilde{\mathbf{p}} + 2wM(\tilde{\mathbf{p}} - \mathbf{p}) \stackrel{!}{=} 0$$

$$\Rightarrow \underline{\left(L^T M L + wM\right)\tilde{\mathbf{p}} = wM\mathbf{p}}$$

compare with implicit Euler!

$NB:$

$$L = M^{-1}L_w \ \Rightarrow\ L^T M L = L_w M^{-1} L_w$$

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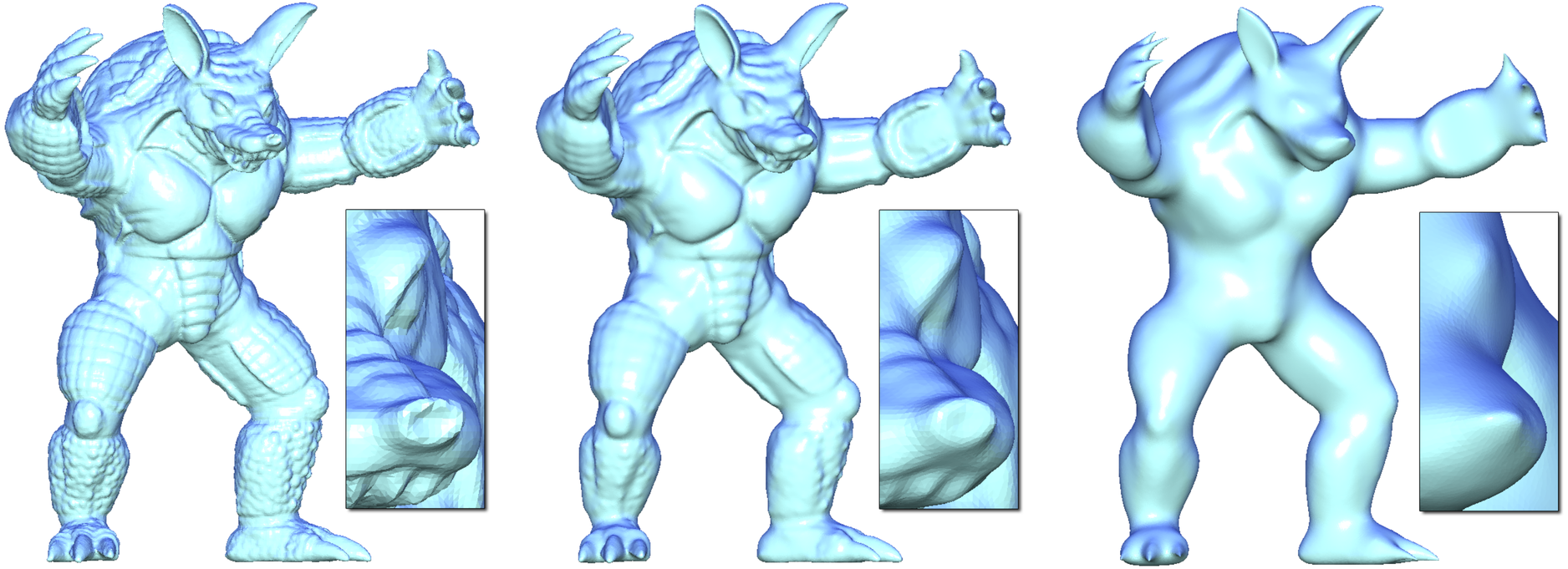
Results



original $w = 0.2$ $w = 0.02$

Customize the energy functional

$$\min_{\tilde{\mathbf{p}}} \sum_{i=1}^{n} A_i \left(\|L\tilde{\mathbf{p}}_i\|^2 + w\|\tilde{\mathbf{p}}_i - \mathbf{p}_i\|^2 \right)$$

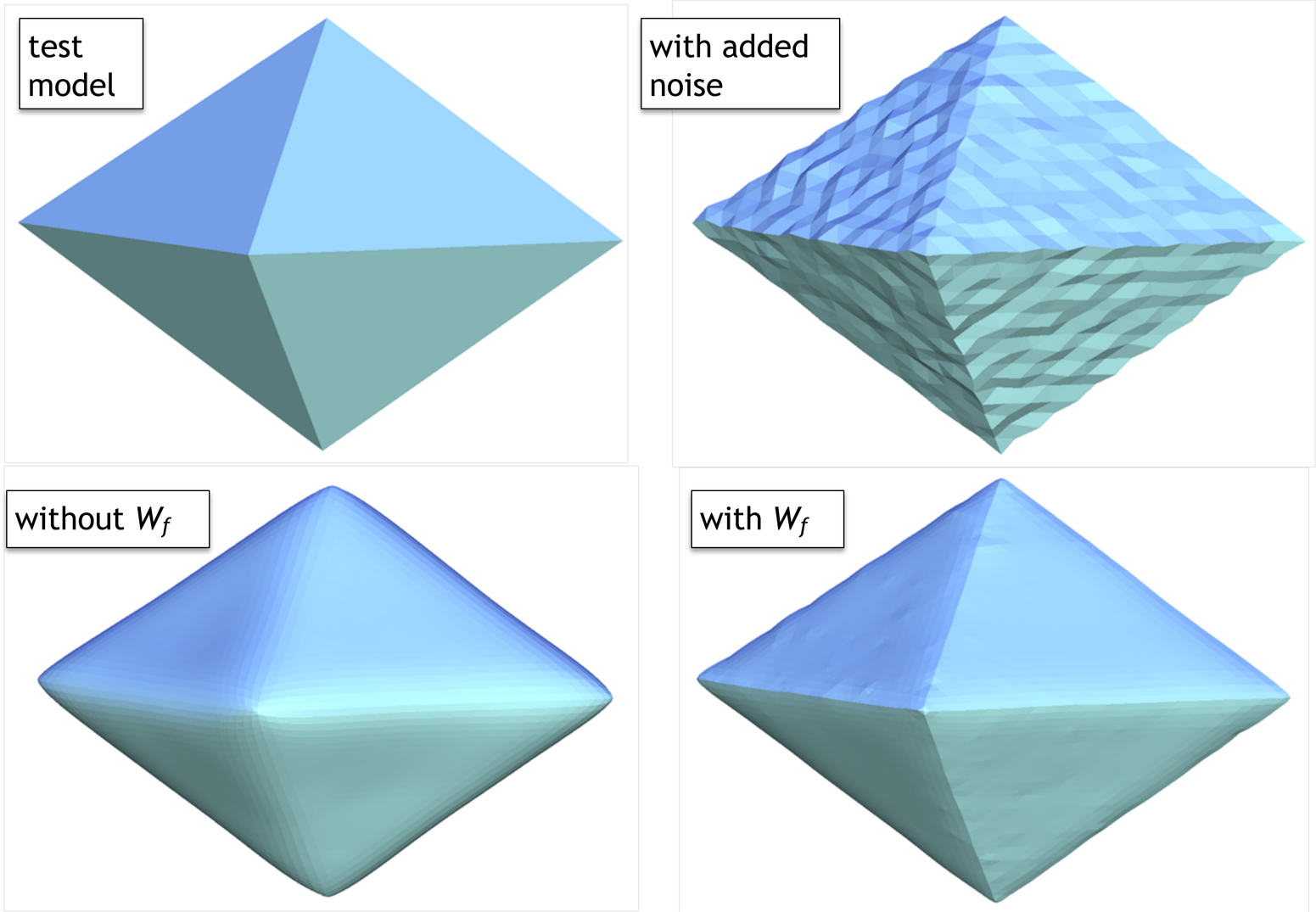
add a varying weight
here to control
feature preservation

$$E(\tilde{\mathbf{p}}) = \tilde{\mathbf{p}}^T L^T (M^{0.5} \underline{W_f} M^{0.5}) L\tilde{\mathbf{p}} + w(\tilde{\mathbf{p}} - \mathbf{p})^T M (\tilde{\mathbf{p}} - \mathbf{p})$$

$(W_f)_{ii}$ can be inverse-proportional to e.g. curvature at vertex *i*

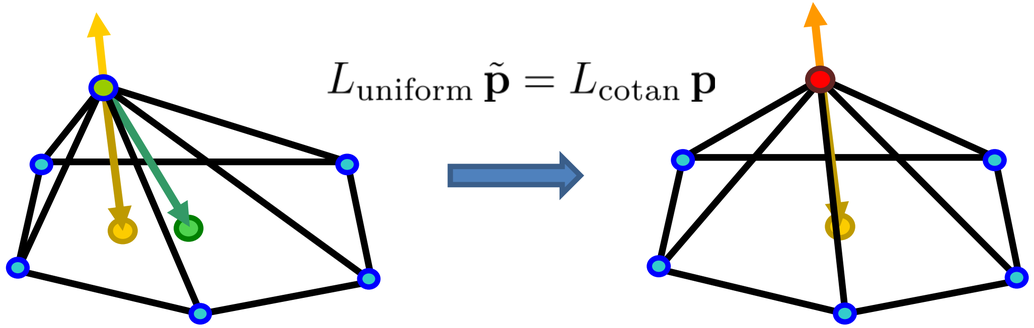
(slight abuse of notation, the energy is
actually the sum of the *xyz* components) $E(\tilde{\mathbf{p}}) = \sum_{\mathbf{v} \in \{\mathbf{x},\mathbf{y},\mathbf{z}\}} \tilde{\mathbf{v}}^T L^T \ldots$

Using W_f



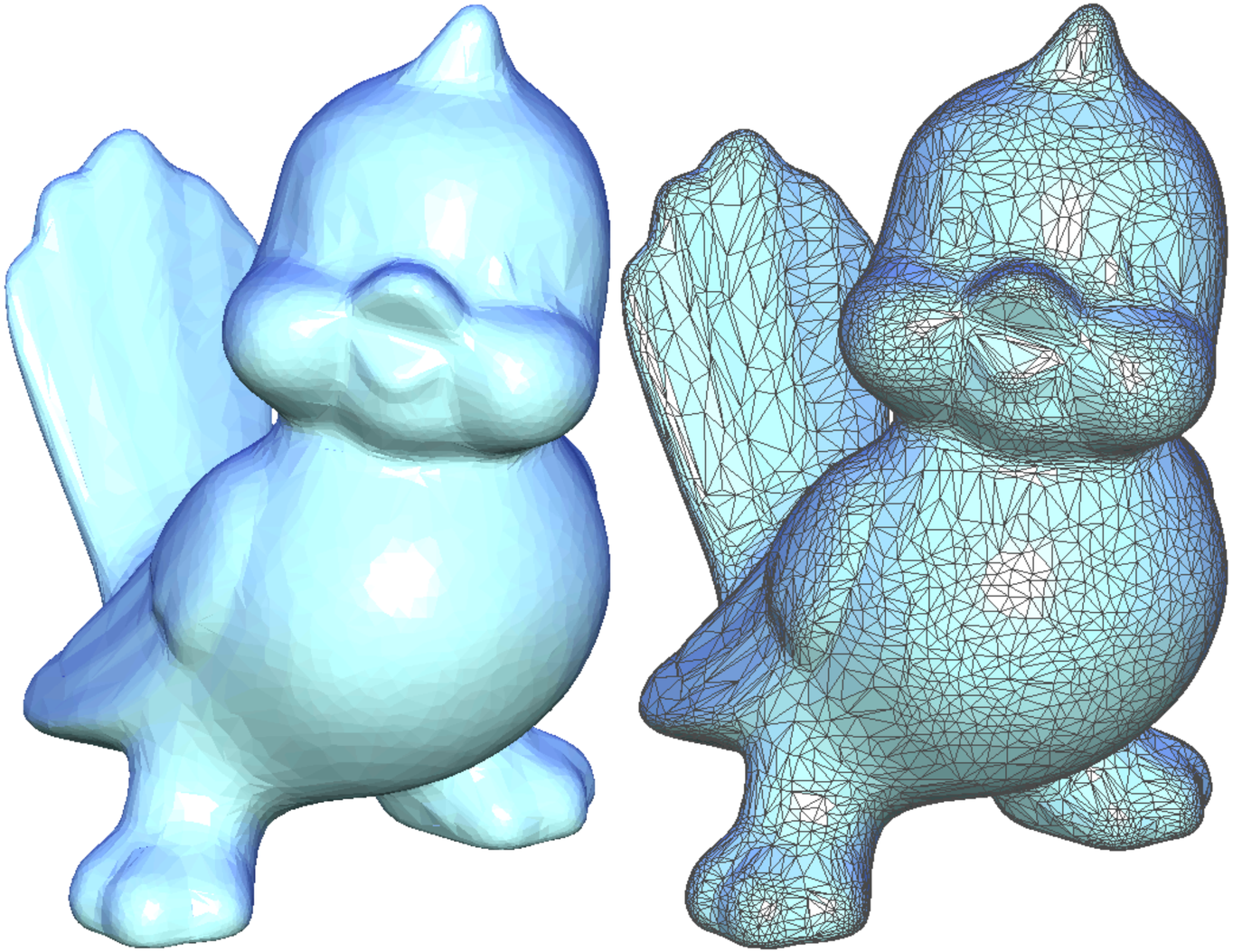
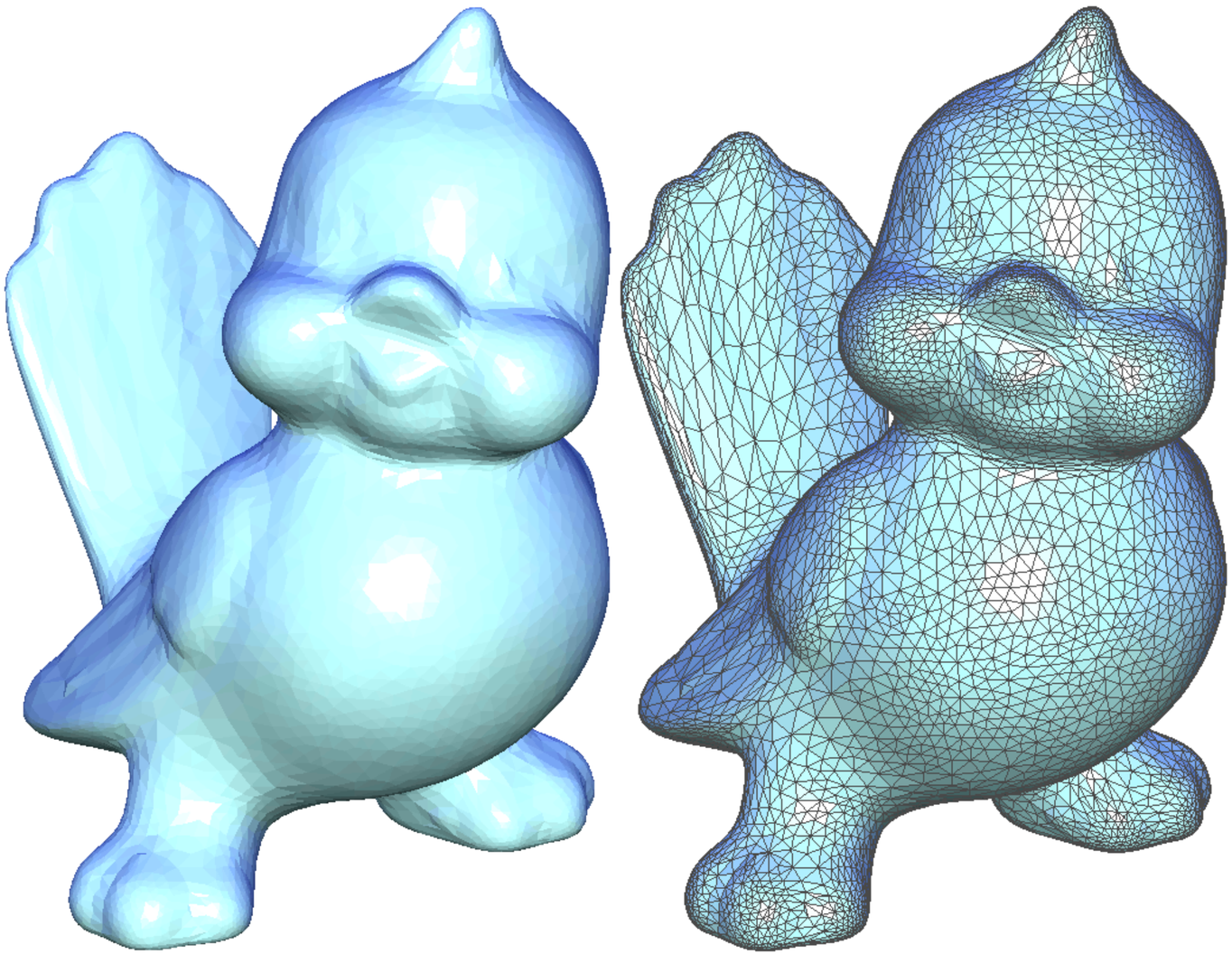
Customize the energy functional

- Can do ***tangential*** smoothing!
  - Improve the shapes of mesh triangles without changing the surface shape



$$\min_{\tilde{\mathbf{p}}} \sum_{i=1}^{n} A_i \left(\| L_{\text{uniform}}\,\tilde{\mathbf{p}}_i - L_{\text{cotan}}\,\mathbf{p}_i \|^2 + w \| \tilde{\mathbf{p}}_i - \mathbf{p}_i \|^2 \right)$$

Original



Triangle Shape Optimization